

ORIGINAL RESEARCH ARTICLE

Evolocumab in Patients With Prior Percutaneous Coronary Intervention and No Prior Myocardial Infarction: Results From the VESALIUS-CV Trial

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BACKGROUND: The clinical benefit of intensive LDL cholesterol (LDL-C) lowering with evolocumab in patients with prior percutaneous coronary intervention (PCI) but without a prior myocardial infarction (MI) is not established.

METHODS: VESALIUS-CV (The Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke) randomized patients with atherosclerosis or high-risk diabetes but without prior MI or stroke and with LDL-C ≥ 90 mg/dL to evolocumab versus placebo. The median follow-up was 4.6 years. The dual primary end points were coronary heart disease death, MI, or ischemic stroke (3-point major adverse cardiovascular event [MACE]) and the same composite plus ischemia-driven revascularization (4-point MACE). For this prespecified subgroup analysis, patients were categorized by whether they had undergone PCI at any time before trial enrollment.

RESULTS: Among 12 257 randomized patients, 3627 (29.6%) had undergone prior PCI with a median time between PCI and enrollment of 4 years. Their median age was 66 years, and 30.7% were women. The median LDL-C in a lipid substudy at 48 weeks was 41.5 (26.0–67.0) mg/dL versus 107.0 (84.0–135.0) mg/dL in the evolocumab versus placebo arms ($P < 0.0001$). Evolocumab reduced the relative rate of 3-point MACE by 30% (5-year Kaplan-Meier rates 7.0% versus 9.5%; hazard ratio [HR], 0.70 [95% CI, 0.56–0.89]; $P = 0.004$) and 4-point MACE by 18% (17.9% versus 21.7%; HR, 0.82 [95% CI, 0.71–0.96]; $P = 0.012$) as well as both MI by 50% (3.0% versus 6.1%; HR, 0.50 [95% CI, 0.36–0.70]; $P < 0.001$), with the effect apparent as soon as 6 months after randomization, and urgent coronary revascularization by 39% (HR, 0.61 [95% CI, 0.46–0.80]; $P < 0.001$). There were nominally lower rates of cardiovascular death (2.6% versus 3.7%; HR, 0.66 [95% CI, 0.45–0.96]; $P = 0.030$) and all-cause death (8.2% versus 10.2%; HR, 0.76 [95% CI, 0.60–0.95]; $P = 0.016$) with evolocumab.

CONCLUSIONS: Evolocumab reduced the risk of major cardiovascular events in stable patients with prior PCI but no MI. These findings support intensive LDL-C lowering in patients who have undergone PCI even in the absence of prior MI.

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Key Words: LDL ■ PCSK9 inhibitor ■ percutaneous coronary intervention

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Clinical Perspective

What Is New?

- The clinical benefit of intensive low-density lipoprotein cholesterol (LDL-C) lowering with evolocumab in patients with prior percutaneous coronary intervention but without a prior myocardial infarction (MI) is not established.
- This was a prespecified subgroup analysis of patients with prior percutaneous coronary intervention from the VESALIUS-CV trial (The Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke).
- Evolocumab reduced the relative rate of 3-point major adverse cardiovascular events by 30% (hazard ratio [HR], 0.70 [95% CI, 0.56–0.89]; $P=0.004$) and 4-point major adverse cardiovascular events by 18% (HR, 0.82 [95% CI, 0.71–0.96]; $P=0.012$) and reduced the relative rate of MI by 50% (HR, 0.50 [95% CI, 0.36–0.70]; $P<0.001$).
- There were nominally lower rates of cardiovascular death (HR, 0.66 [95% CI, 0.45–0.96]; $P=0.030$) and all-cause death (HR, 0.76 [95% CI, 0.60–0.95]; $P=0.016$) with evolocumab.

What Are the Clinical Implications?

- In this prespecified subgroup analysis from the VESALIUS-CV trial, we found that evolocumab substantially reduced the risk of major cardiovascular events in stable patients with prior percutaneous coronary intervention but no prior MI or stroke.
- The benefits were consistent across a range of cardiovascular outcomes, with lower rates seen with evolocumab for the primary composite end points and coronary-specific outcomes such as MI, as well as nominally lower rates of cardiovascular and all-cause mortality.
- These findings support intensive LDL-C lowering in patients who have undergone percutaneous coronary intervention even in the absence of prior MI.

The clinical importance of intensive LDL cholesterol (LDL-C)-lowering in high-risk patients with a prior myocardial infarction (MI) or other atherosclerotic cardiovascular event is firmly established.^{1,2} The VESALIUS-CV trial (The Effect of Evolocumab in Patients at High Cardiovascular Risk Without Prior Myocardial Infarction or Stroke) showed marked reductions in clinical risk with the PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitor evolocumab in patients with qualifying established atherosclerosis or high-risk diabetes but without prior MI or stroke, significantly broadening the population that stands to benefit from this therapy.³

Included in the VESALIUS-CV trial were patients with prior percutaneous coronary intervention (PCI) in the absence of MI.^{3,4} These are important patients to study, as they may be at heightened risk for cardiovascular

Nonstandard Abbreviations and Acronyms

CHD	coronary heart disease
IDR	ischemia-driven revascularization
IQR	interquartile range
LDL-C	low-density lipoprotein cholesterol
MACE	major adverse cardiovascular events
MI	myocardial infarction
PCI	percutaneous coronary intervention
STEMI	ST-segment-elevation myocardial infarction

events and are commonly encountered in clinical practice. Additionally, patients with stable atherosclerosis, including those undergoing PCI in the absence of MI, have low rates of LDL-C measurement and optimization in clinical practice.^{5,6} The effect of evolocumab on clinical events in these patients has not been described.

METHODS

The VESALIUS-CV trial database cannot be shared, but investigators interested in collaboration should contact the corresponding author. The design and primary findings of the VESALIUS-CV trial (NCT03872401) have been published previously.^{3,4} VESALIUS-CV was a randomized, placebo-controlled, multinational trial of subcutaneously administered evolocumab (140 mg every 2 weeks) in patients with established atherosclerosis or high-risk diabetes but without prior MI or stroke. Patients were required to be on optimized lipid-lowering therapy and have an LDL-C level of at least 90 mg/dL, a non-HDL-C level of at least 120 mg/dL, or an apolipoprotein B level of at least 80 mg/dL. Complete inclusion and exclusion criteria, including definitions of established atherosclerosis and high-risk diabetes, have been published.^{3,4} In this prespecified subgroup analysis, we examined outcomes in patients who had undergone PCI previously. Prior PCI status was documented dichotomously by sites at the time of patient screening without further granularity regarding the number of prior interventions. Written informed consent was obtained from all patients, and approval was obtained from all relevant ethics review committees.

The dual primary end points for the trial were coronary heart disease (CHD) death, MI, or ischemic stroke (3-point major adverse cardiovascular event [MACE]) and the same composite plus ischemia-driven revascularization (4-point MACE). Ischemia-driven revascularization (IDR) was defined as any arterial revascularization in the presence of ischemia affecting the relevant end organ and could involve coronary, cerebrovascular, or peripheral arteries. Secondary efficacy end points included the following 4 composites: (1) MI, ischemic stroke, or IDR; (2) CHD death, MI, or IDR; (3) cardiovascular death, MI, or ischemic stroke; and (4) CHD death or MI, and the individual end points of MI, IDR, CHD death, cardiovascular death, all-cause mortality, and ischemic stroke. Urgent revascularization events were defined as those where the procedure was clinically indicated to be performed as soon as possible or on an

inpatient basis before discharge. All potential clinical end point events were reviewed by a central clinical events committee blinded to treatment arm and lipid levels. Follow-up lipid levels were not collected in all patients; however, a subset of patients from participating sites was included in the lipid substudy for prespecified analyses, including the change in LDL-C from baseline in the evolocumab arm compared with placebo.

Statistical Analysis

Both primary and secondary efficacy analyses were conducted on all patients who were randomized (intention-to-treat method). For the lipid substudy, analyses were performed on all patients who had been randomized, participated in the substudy, received at least 1 dose of evolocumab or placebo, and had at least 1 postbaseline lipid measurement analyzed by the central laboratory. The changes in LDL-C and other lipid parameters were reported as placebo-adjusted least squares means using a mixed model with repeated measures, including treatment arm, study visit, interaction of treatment arm and study visit, and protocol-specified stratification factors. The covariance structure type used for repeated measures was unstructured, and for the analyses of changes from baseline to follow-up time points there were no adjustments for baseline level as prespecified in the original trial statistical analysis plan. Event rates are reported as 5-year Kaplan-Meier percentages. Time-to-event analyses were performed using the log-rank test, with stratification by screening LDL-C and geographic region. In addition, hazard ratios (HRs) with 95% CIs were estimated from a Cox model, stratified as above. The proportional hazards assumption was assessed for each end point by examining plots of scaled Schoenfeld residuals on functions of time and tests of whether the association is time dependent. Heterogeneity testing was performed on the basis of prior PCI status. A post hoc sensitivity analysis to address the competing risk of death was performed for the 2 primary end points using a Fine-Gray model including any death other than the specific type of death included in each end point as a competing risk. Statistical analyses were performed using SAS version 9.4 (SAS Institute Inc, Cary, NC). The VESALIUS-CV trial database cannot be shared, but investigators interested in collaboration should contact the corresponding author.

RESULTS

Among the 12 257 randomized patients, 3627 (29.6%) had undergone prior PCI (Table). The median age was 66 years (interquartile range [IQR], 60–71), and 30.7% were women. The median time from the most recent PCI to trial enrollment was 4 (IQR, 1–8) years. A total of 76.6% were on a high-intensity lipid-lowering regimen, including 71.4% taking a high-intensity statin; 80.3% were taking aspirin, and 31.3% were on dual antiplatelet therapy. Characteristics were generally similar across treatment arms. Patients with prior PCI compared with those without prior PCI were less likely to be women, less likely to have diabetes, and more likely to be on a high-intensity statin and ezetimibe, with lower LDL-C at baseline (Table S1).

In the placebo arm, compared with patients without a prior PCI, patients with a prior PCI had higher rates of

Table. Baseline Characteristics by Treatment Arm in Patients With Prior PCI

	Evolocumab (n=1820)	Placebo (n=1807)
Age, y	66 (60–71)	66 (60–71)
Women	562 (30.9)	553 (30.6)
White patients	1640 (90.2)	1625 (90.0)
Region		
North America	183 (10.1)	185 (10.2)
Europe	1291 (70.9)	1301 (72.0)
Asia/Pacific	156 (8.6)	145 (8.0)
Latin America	190 (10.4)	176 (9.7)
Hypertension	1624 (89.2)	1613 (89.3)
Diabetes	803 (44.1)	748 (41.4)
High-risk diabetes	562 (30.9)	506 (28.0)
Current smoker	472 (25.9)	439 (24.3)
Family history of premature CAD	439 (24.1)	451 (25.0)
Menopause before 40 y	35/562 (6.2)	40/553 (7.2)
Time from most recent PCI, y	4 (1–8)	4 (1–8)
Prior CABG	228 (12.5)	241 (13.3)
Lipoprotein(a) >50 mg/dL (125 nmol/L)	86 (21.6)	80 (20.5)
eGFR <60 mL/min per 1.73 m ²	342 (18.8)	367 (20.4)
Any statin	1615 (88.7)	1615 (89.4)
High-intensity statin	1288 (70.8)	1300 (71.9)
Ezetimibe	462 (25.4)	469 (26.0)
High-intensity lipid-lowering regimen	1375 (75.5)	1404 (77.7)
Aspirin	1452 (79.8)	1459 (80.7)
Dual antiplatelet therapy	568 (31.2)	567 (31.4)
LDL-C, mg/dL	116.0 (100.5–140.4)	115.0 (101.3–139.2)
Non-HDL-C, mg/dL	146.9 (126.1–175.6)	144.6 (125.7–173.2)
Apolipoprotein B, mg/dL	98.0 (86.0–117.0)	95.0 (84.0–110.0)

Continuous variables are presented as median (interquartile range) and counts as n (percent). Only patients with available data are included in the denominators for percent calculations. Baseline lipid levels are among patients in the lipid substudy. CABG indicates coronary artery bypass graft surgery; CAD, coronary artery disease; eGFR, estimated glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; PCI, percutaneous coronary intervention.

both primary end points: 5-year Kaplan-Meier event rates of 9.5% versus 7.3% (HR, 1.43 [95% CI, 1.18–1.74]; $P<0.001$) for 3-point MACE and 21.7% versus 13.9% (HR, 1.66 [95% CI, 1.45–1.90]; $P<0.001$) for 4-point MACE (Table S2). In terms of coronary events, patients with a prior PCI had about twice the rate of CHD death or MI (7.7% versus 4.7%; HR, 1.80 [95% CI, 1.44–2.25]; $P<0.001$), MI (6.1% versus 3.3%; HR, 1.97 [95% CI, 1.51–2.56]; $P<0.001$), and urgent coronary IDR (8.2% versus 3.4%; HR, 2.62 [95% CI, 2.05–3.34]; $P<0.001$) compared with patients without prior PCI.

A total of 507 patients in the prior PCI subgroup had a baseline lipid panel as part of the lipid sub-study, of whom 433 (85.4%) and 401 (79.1%) had 48-week and 96-week laboratory tests, respectively. The median baseline LDL-C level was 109.5 (IQR, 93.0–138.0) mg/dL in the evolocumab arm and 105.0 (IQR, 89.0–131.0) mg/dL in the placebo arm. After 48 weeks, treatment with evolocumab resulted in a least squares mean 71.7 mg/dL reduction in LDL-C (95% CI, –81.6, –61.7) and a median achieved LDL-C level of 41.5 (IQR, 26.0–67.0) mg/dL compared with 107.0 (IQR, 84.0–135.0) mg/dL in the placebo group (Figure 1; Tables S3 and S4). This resulted in a 66.4 (95% CI, 59.0–73.9) mg/dL difference in LDL-C between the evolocumab and placebo groups. After 96 weeks, the corresponding median achieved LDL-C levels were 40.0 (IQR, 21.0–63.0) mg/dL and 97.0 (IQR, 80.0–125.0) mg/dL, respectively. There were also significant reductions in non-HDL-C and apolipoprotein B with evolocumab (Tables S3 and S4).

In patients with prior PCI, evolocumab reduced the relative rate of 3-point MACE by 30% (7.0% versus 9.5%; HR, 0.70 [95% CI, 0.56–0.89]; $P=0.004$) and 4-point MACE by 18% (17.9% versus 21.7%; HR, 0.82 [95% CI, 0.71–0.96]; $P=0.012$) (Figure 2). Evolocumab demonstrated consistent benefits across the prespecified secondary end points (Figure 3), including a reduction in MI by 50% (3.0% versus 6.1%; HR, 0.50 [95% CI, 0.36–0.70]; $P<0.001$), coronary IDR by 26% (11.6% versus 15.7%; HR, 0.74 [95% CI, 0.61–0.89]; $P=0.002$),

and urgent coronary IDR by 39% (4.8% versus 8.2%; HR, 0.61 [95% CI, 0.46–0.80]; $P<0.001$). The reduction in MI reached statistical significance as early as 6 months after randomization (0.3% versus 0.8%; HR, 0.33 [95% CI, 0.12–0.90]; $P=0.030$) (Figure 4) and through the full trial follow-up included a 60% reduction in ST-segment-elevation MI (STEMI) (0.6% versus 1.7%; HR, 0.40 [95% CI, 0.21–0.79]; $P=0.008$) and a 45% reduction in non-STEMI (2.3% versus 4.2%; HR, 0.55 [95% CI, 0.37–0.82]; $P=0.003$). Evolocumab did not have an apparent effect on the incidence of ischemic stroke in the prior PCI subgroup (2.9% versus 2.4%; HR, 1.06 [95% CI, 0.69–1.64]; $P=0.79$). There were nominally lower rates of cardiovascular death (2.6% versus 3.7%; HR, 0.66 [95% CI, 0.45–0.96]; $P=0.030$) and all-cause death (8.2% versus 10.2%; HR, 0.76 [95% CI, 0.60–0.95]; $P=0.016$) with evolocumab.

The effects of evolocumab on the primary end points and on MI were consistent regardless of the time between PCI and trial enrollment (Figure S1). Evolocumab's effects were also generally consistent for other clinical subgroups among the patients with prior PCI (Tables S5 and S6). Rates of MACE were largely consistent in the placebo arm regardless of whether patients had more recent (<1 year, $n=286$) versus more distant (≥ 1 year, $n=1520$) prior PCI. For 3-point MACE, the Kaplan-Meier incidence rate was 7.5% for those with prior PCI within 1 year of enrollment compared with 9.8% for those with a longer interval since PCI. For 4-point MACE, these rates were 22.4% and 21.5%,

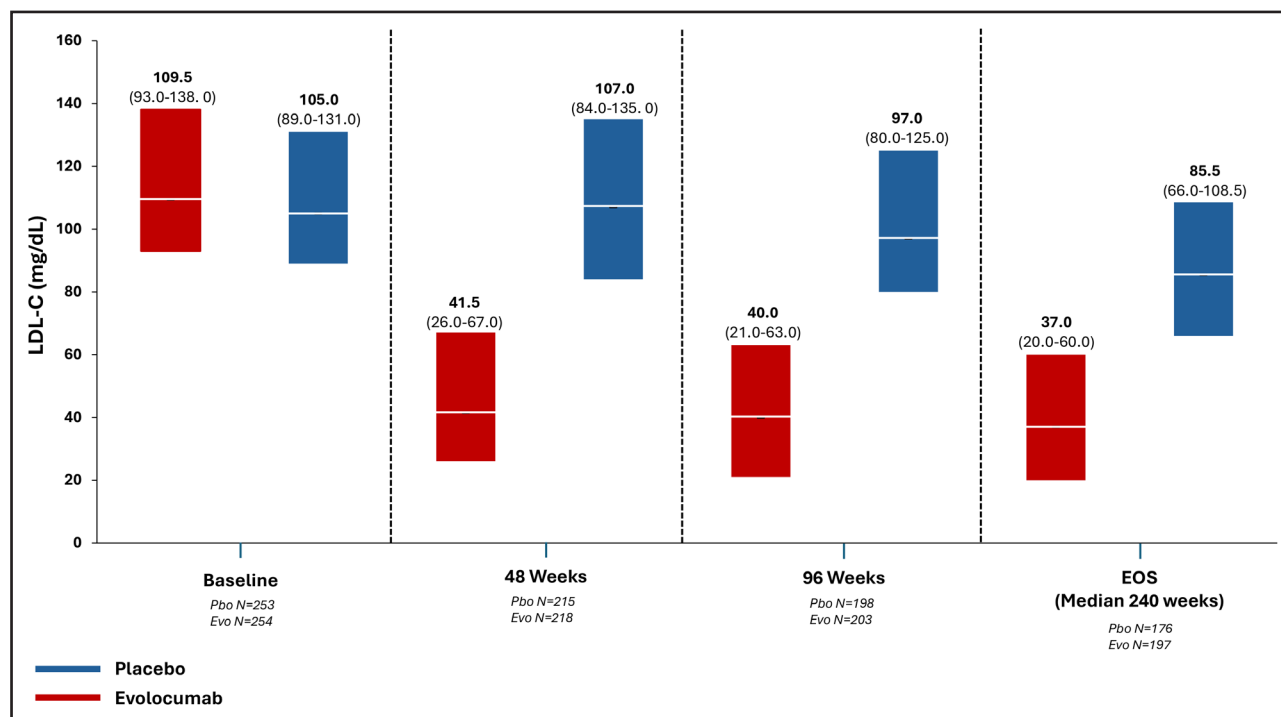


Figure 1. Low-density lipoprotein cholesterol (LDL-C) level by treatment arm from baseline through end of study (EOS) in patients with prior percutaneous coronary intervention enrolled in the lipid sub-study.

Values shown are median and interquartile range. Evo indicates evolocumab and Pbo, placebo.

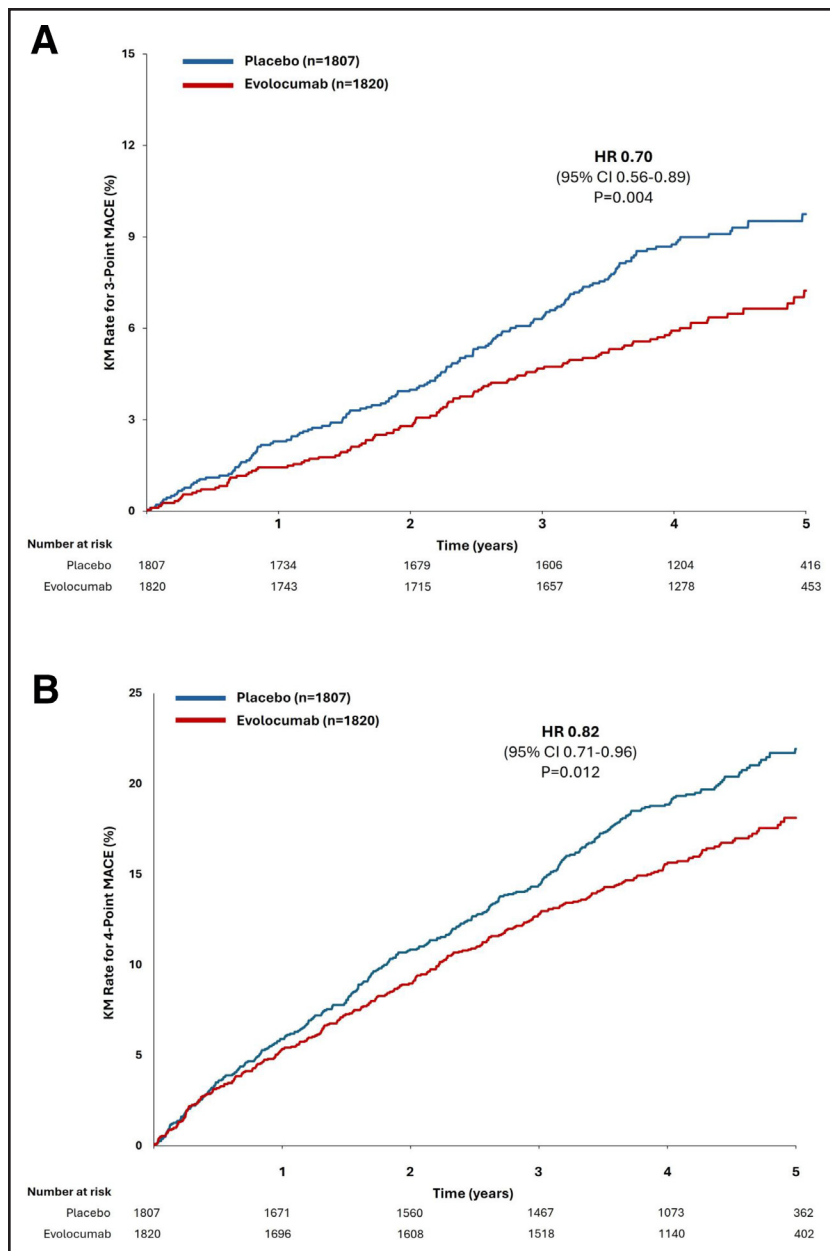


Figure 2. Effect of evolocumab on the dual primary end points of 3-point major adverse cardiovascular event (MACE) (A) and 4-point MACE (B) in patients with prior percutaneous coronary intervention.

The 3-point MACE composite included coronary heart disease death, myocardial infarction, or ischemic stroke; the 4-point MACE composite included these same end points plus ischemia-driven revascularization. HR indicates hazard ratio and KM, Kaplan-Meier.

respectively. The effect of evolocumab on the primary end points was also consistent in the competing risk analysis (3-point MACE: subdistribution HR, 0.71 [95% CI, 0.56–0.90]; 4-point MACE: subdistribution HR, 0.83 [95% CI, 0.71–0.96]).

The relative reductions in the dual primary end points seen with evolocumab were similar in patients with versus without prior PCI; for 3-point MACE, the HRs were 0.70 (95% CI, 0.56–0.89) and 0.77 (95% CI, 0.65–0.92) ($P_{\text{int}}=0.53$), and for 4-point MACE, the HRs were 0.82 (95% CI, 0.71–0.96) and 0.79 (95% CI, 0.70–0.90) ($P_{\text{int}}=0.75$) (Table S7). However, for coronary events, the magnitude of clinical benefit tended to be greater with evolocumab among patients with prior PCI, with a 42% reduction (HR, 0.58 [95% CI, 0.44–0.76]) in CHD death or MI as compared with a 16% reduction (HR, 0.84 [95%

CI, 0.68–1.04]) in patients without prior PCI ($P_{\text{int}}=0.036$). Similar patterns were seen for CHD death (HR, 0.69 [95% CI, 0.43–1.09] versus 1.01 [95% CI, 0.73–1.39]), MI (HR, 0.50 [95% CI, 0.36–0.70] versus 0.75 [95% CI, 0.58–0.98]) and urgent coronary IDR (HR, 0.61 [95% CI, 0.46–0.80] versus 0.74 [95% CI, 0.57–0.96]).

There were no significant differences in the rates of adverse events, serious adverse events, or adverse events leading to study drug discontinuation across treatment arms in patients with or without prior PCI (Table S8).

DISCUSSION

In this prespecified subgroup analysis from the VESALIUS-CV trial, we found that evolocumab substantially reduced the risk of major cardiovascular events in

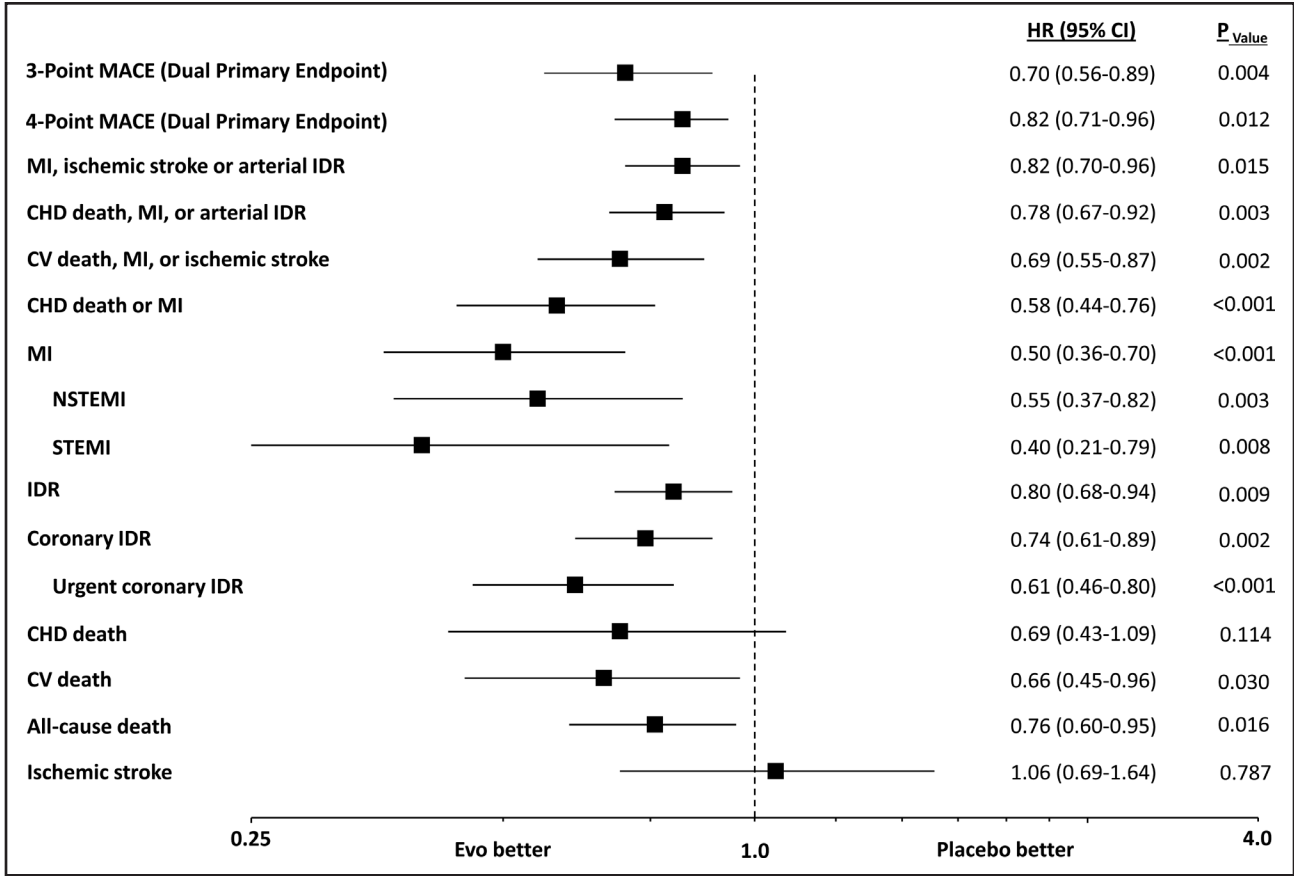


Figure 3. Effect of evolocumab on the primary and secondary end points in patients with prior percutaneous coronary intervention.

The 3-point major adverse cardiovascular event (MACE) composite included coronary heart disease death, myocardial infarction, or ischemic stroke; the 4-point MACE composite included these same end points plus ischemia-driven revascularization. CHD indicates coronary heart disease; CV, cardiovascular; Evo, evolocumab; HR, hazard ratio; IDR, ischemia-driven revascularization; MI, myocardial infarction; NSTEMI, non–ST-segment-elevation myocardial infarction; and STEMI: ST-segment-elevation myocardial infarction.

stable patients with prior PCI but no prior MI or stroke. The benefits were consistent across a range of cardiovascular outcomes, with lower rates seen with evolocumab for the primary composite end points and coronary-specific outcomes such as MI as well as nominally lower rates of cardiovascular and all-cause mortality.

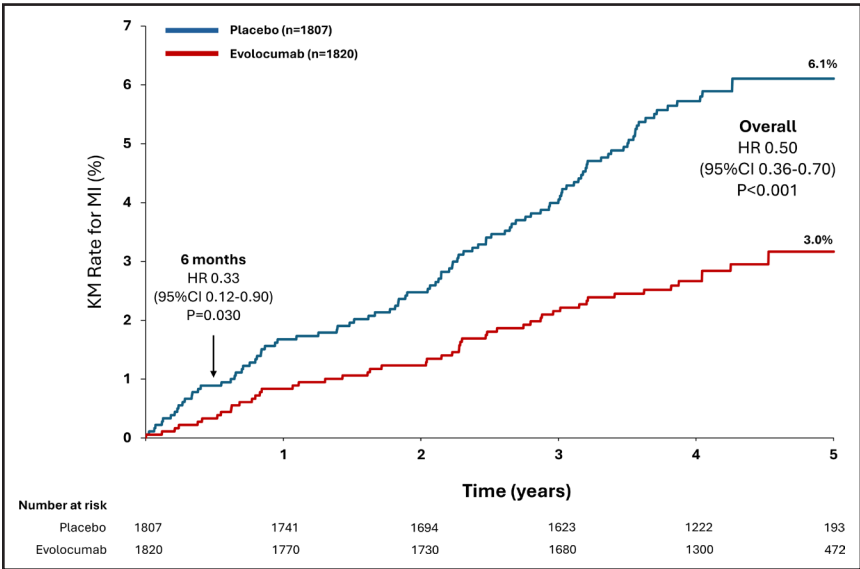


Figure 4. Incidence of myocardial infarction by treatment arm in patients with prior percutaneous coronary intervention.

HR indicates hazard ratio; KM, Kaplan-Meier; and MI, myocardial infarction.

Patients with prior coronary intervention but without a history of an acute atherosclerotic event such as MI or stroke are a common and clinically important population, with approximately one-third to one-half of PCIs in the United States performed outside the context of an MI.⁷ In the classical paradigm, these patients have not been considered to be at high clinical risk for events by virtue of not having had such an event previously.^{2,8,9} We observed in VESALIUS-CV that these patients with prior PCI in the placebo arm were indeed at heightened risk for numerous types of adverse outcomes, including an approximately 2-fold higher rate of coronary-specific events such as MI, including both non-STEMI and STEMI. In fact, the event rates in this prior PCI subgroup were similar to those seen in the secondary prevention FOURIER and ODYSSEY-OUTCOMES populations.^{1,10} The higher cardiovascular risk seen among the prior PCI patients compared with those without prior PCI occurred despite more frequent use of high-intensity statin, ezetimibe, aspirin, and dual antiplatelet therapy use at baseline, as would be expected, and lower baseline LDL-C levels. In this group with elevated absolute event rates, evolocumab treatment resulted in substantial relative reductions, including 30% and 18% reductions in the primary end points, a 50% reduction in MI with a 60% reduction in STEMI, and nominal 34% and 24% reductions in cardiovascular death and all-cause mortality, respectively.

These findings extend the benefits previously seen in the higher-risk patients studied in the FOURIER and ODYSSEY-OUTCOMES trials to a broader, pre-event population.^{1,10–12} PCSK9 inhibition has been shown to have favorable effects on coronary plaque volume and composition following an MI^{13,14} as well as in stable patients with atherosclerosis.¹⁵ Additionally, we previously showed in an analysis from the FOURIER trial, that patients treated with evolocumab had significantly lower rates of complex PCI or coronary bypass surgery than patients in the placebo arm and had less complex coronary anatomy at the time of revascularization.¹¹ The VESALIUS-CV findings indicate that these clinical benefits seen in higher-risk patients and the surrogate coronary imaging findings translate into meaningful reductions in coronary events in patients with prior PCI but without prior MI.

While recent guidelines have begun to support more intensive LDL-C lowering in patients with established atherosclerotic cardiovascular disease but without a prior acute event,^{2,16} in clinical practice, patients following PCI outside the MI setting have been shown to have low rates of LDL-C monitoring and optimization.^{5,6} Whereas post-MI care oftentimes follows established clinical pathways and metrics, management of stable atherosclerotic cardiovascular disease is more varied. Additionally, numerous clinical, logistical, and financial barriers to medical optimization persist for many patients.¹⁷ Therefore, the

large difference in LDL-C levels achieved here with evolocumab compared with placebo reflects the potential benefit in clinical practice with intensive optimization of LDL-C to approximately 40 mg/dL and supports an early and aggressive approach in these patients.¹⁸

Importantly, the clinical benefits of evolocumab manifested as early as 6 months and were consistent irrespective of the time interval between PCI and trial enrollment. These observations indicate that early initiation of evolocumab is beneficial, but patients still stand to gain even if identified quite late post PCI. There was no effect of evolocumab on the risk of ischemic stroke in this subgroup. There are multiple potential etiologies for ischemic stroke, only some of which are modifiable with lipid-lowering therapy. Moreover, there were a small number of strokes in this subgroup and therefore very wide CIs that overlap the effect seen in the overall trial (HR, 0.79 [95% CI, 0.62–1.01]).

This study has limitations. While the timing of PCI before trial enrollment was documented, the specific coronary anatomy and details of the prior intervention were not known. Therefore, whether there is heterogeneity based on the indication for prior intervention cannot be discerned. Similarly, for incident events during the trial, specific data regarding the target lesion and vessel were not available in case report forms. For less frequent event types, the numbers of events per group were small once patients were divided by prior PCI status and treatment arm, limiting the ability to detect differences. Finally, trial enrollment required LDL-C to be at least 90 mg/dL on optimized therapy, and these findings might not be assumed to apply equally to patients with lower LDL-C levels. However, the magnitude of clinical benefit per unit reduction in LDL-C in this trial was consistent with that seen for statin trials in the Cholesterol Treatment Trialists Collaboration. These observations suggest that the benefit is proportional to the LDL-C reduction all the way down to levels of approximately 40 mg/dL.

In conclusion, evolocumab rapidly and substantially reduced the risk of major cardiovascular events in stable patients with prior PCI but no MI. These findings support intensive LDL-C lowering in patients who have undergone PCI even in the absence of prior MI.

ARTICLE INFORMATION

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Disclosures

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Supplemental Material

Tables S1–S8
Figure S1

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